Explicit Hodge Classes for a Fixed CM Field

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Abstract

We present explicit algebraic-cycle and character-lattice arguments for products of CM abelian varieties attached to the fixed field $\mathbb{Q}(\zeta_{42})$. Integral relation generation, algebraic source tensors and contractions yield the stated mixed-product algebraicity results for its eight rank-one sixfold type orbits. The note records the proof inputs used by the packet and quotient papers and distinguishes rational Hodge conclusions from integral polarizations and curve placement. It is a fixed-field supporting calculation, not a claim of a new general CM Hodge theorem or a verification of a proposed general one.

Companion references. Result numbers prefixed I, II, H and S refer respectively to the packet paper, quotient paper, supporting Hodge note and computational supplement. Each document has its own numbering and bibliography.

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1 Purpose and notation

This supporting note records explicit Hodge-algebraicity arguments for the fixed field $M = \mathbb{Q}(\zeta_{42})$. The strongest statement is Theorem 4.1: arbitrary finite mixed products of the eight rank-one M CM-type orbits. The earlier mixed-power and six-orbit results are retained as proof stages, rather than presented as separate novelty claims. They supply source tensors used in the final argument.

A rank-one M sixfold means a complex abelian sixfold whose rational first cohomology has dimension one over M. Embeddings are labelled by units modulo 42. We use $K = \mathbb{Q}(\sqrt{-3})$, $L = M^+$ and $\Phi_B = \{1, 5, 23, 25, 29, 31\}$. The symbols A_A, A_B, A_C in the packet corollary refer to the geometric fibers of Paper I, Theorem I.1.2. That application uses only their stated isogeny classes; the abstract mixed-product theorem does not rely on the packet intersection search.

The route is: explicit algebraic input cycles, a saturated integral character-relation lattice, tensor products and contractions, and rational descent. A rational relation basis alone would not justify the integral step. Computational relation checks supplement the arguments; they do not by themselves prove cycle algebraicity.

OpenAI's proposed general CM theorem [7], if valid, implies the conclusions here. André's reduction and the split-Weil literature [1, 6, 3, 9], and the Fermat antecedents discussed by Saller [8], must be included in assessing novelty. This note claims neither a new general CM range nor an independent verification of the OpenAI theorem. The reported Aoki route is not an adopted proof input. Its role is to document the explicit fixed-field construction supporting the two arithmetic-geometric papers.

Theorem 2.1 is proved by the algebraic input construction and the synchronized relation calculation of Sections 2.2–2.4. The individual-powers statement follows from the eight-type analysis following that calculation; the last section proves the full mixed result.

2 Hodge algebraicity on mixed powers

2.1 The mixed-powers and individual-powers statements

Put $M = \mathbb{Q}(\zeta_{42}) = \mathbb{Q}(\zeta_{21})$. Embeddings of M are labeled by units modulo 42. In this section, B is the underlying complex isogeny class of A_B with a relabeled M-action, and D_{CM} is an additional abstract sixfold. The notation D_{CM} is unrelated to the quadrilateral generator D_t or its vertex, which is equivalent to B_t. We use Ψ, rather than the geometric types Φ of Paper I, for the following rank-one M CM types:

$$\Psi_B = \{1, 5, 11, 13, 17, 19\}, \qquad \Psi_{D_{\mathrm{CM}}} = \{1, 5, 11, 13, 19, 25\}.$$

In terms of the geometric type Φ_B just specified, $\Psi_B = 19\Phi_B$ modulo 42. This relabeling changes the field marking used in the character calculation, not the underlying complex isogeny class. Here rank one means that rational first cohomology is a one-dimensional M-vector space. All varieties and maps in this section are over $\mathbb{C}$. Let Y be the Jacobian of the smooth projective model of $y^2 = x^7 - 1$.

Theorem 2.1 (Mixed powers)**.** *For all nonnegative integers r, s, t, every rational Hodge class on $B^r \times {D_{\mathrm{CM}}}^s \times Y^t$ is algebraic, in every codimension. The statement is invariant under replacing any factor by a complex isogenous abelian variety.*

Corollary 2.2 (The packet fibers and the additional abstract type)**.** *Let A_A, A_B, A_C be the three certified packet fibers of Theorem* I.1.2*. Then every rational Hodge class on $A_A^a \times A_B^b \times A_C^c \times D_{\mathrm{CM}}^d$ is algebraic for all nonnegative a, b, c, d.*

Proof. The product is isogenous to $Y^{2a} \times B^{b+c} \times {D_{\mathrm{CM}}}^{d}$. An isogeny and its rational inverse act by algebraic correspondences; pullback and pushforward divided by its degree transport algebraicity. The isogeny identifications are Proposition I.5.18. They concern the underlying varieties, not their prescribed polarizations or RM markings. Equal rank-one CM types identify rational Hodge structures, and commensurable full lattices give a complex isogeny after clearing denominators; see also [5, §3.11, Prop. 3.12]. □

Theorem 2.3 (Individual powers for the fixed field)**.** *If X is a complex abelian sixfold with a rational M-action and $\dim_M H^1(X, \mathbb{Q}) = 1$, then every rational Hodge class on every positive power X^n is algebraic.*

Theorem 2.3 ranges over eight CM-type orbits, while Theorem 2.1 concerns the stated three isogeny classes. The later Theorem 4.1 establishes algebraicity on arbitrary mixed products of all eight rank-one M types. The symbol D_{CM} denotes an abstract CM type here; no new curve point or fourth quotient class is asserted. None of these statements identifies a standard trace polarization with a principal polarization, or proves literal membership in a prescribed principally polarized moduli locus.

2.2 The algebraic input classes

For a smooth projective T, write

$$\mathrm{Hdg}^p(T) = H^{2p}(T, \mathbb{Q}) \cap H^{p,p}(T), \qquad \mathrm{Alg}^p(T) = \mathrm{im}\big(\mathrm{CH}^p(T)_{\mathbb{Q}} \longrightarrow H^{2p}(T, \mathbb{Q})\big).$$

For a splitting field E, algebraicity of a split eigenline means containment in $\mathrm{Alg}^p(T) \otimes_{\mathbb{Q}} E$. It does not assert that an individual vector of that line is rational.

2.2.1 The B/Y input

Let $J_m = J(y^2 = x^m - 1)$ for odd m. Shioda's original Example 6.3 proves the rational Hodge conjecture for J_{21} in all codimensions [10, pp. 76–77]. It is a theorem about this Jacobian, not an assertion about all its powers. Its factorization is

$$J_{21} \sim X_{21} \times J_7 \times J_3,$$

where X_{21} is the primitive six-dimensional factor. The primitive holomorphic character set modulo 21 is $\{1, 2, 4, 5, 8, 10\}$; multiplication by 19 gives the restriction of Ψ_B. Thus $X_{21} \sim B$ by CM classification. Projection pullback and restriction to the zero point of J_3, together with isogeny transport, prove algebraicity of all Hodge classes on $B \times Y$. This also supplies algebraicity on B itself. The source audit is retained in `experiments/mixed_products_v100/literature_scope_review.json`.

2.2.2 An explicit split Weil polarization for D_{CM}

Write $\zeta = e^{2\pi i/42}$, $u = \zeta + \zeta^{-1}$, $L = M^+$, and $K_7 = \mathbb{Q}(\sqrt{-7})$. Set

$$s_3 = 1 + 2\zeta^{14} = i\sqrt{3}, \qquad s_7 = \sum_{a=1}^{6} \chi_{-7}(a)\zeta^{6a} = i\sqrt{7}, \qquad u_a = \zeta^a + \zeta^{-a},$$

where the signs are fixed at the displayed complex embedding. Put

$$b = \sqrt{21}\, u_1 u_{13} u_{17} = 4u^5 + 3u^4 - 15u^3 - 13u^2 + 4u + 3,$$
$$\xi = s_7 b, \qquad f(u) = u^6 + u^5 - 6u^4 - 6u^3 + 8u^2 + 8u + 1 = 0.$$

The equality $\sqrt{21} = -s_3 s_7$ fixes its positive sign. The exact sign data at the real embeddings $a = 1, 5, 11, 13, 17, 19$ are

a	1	5	11	13	17	19
ϵ_a for $\Psi_{D_{\mathrm{CM}}}$	+	+	+	+	−	+
$\chi_{-7}(a)$	+	−	+	−	−	−
$\operatorname{sign} \sigma_a(b)$	+	−	+	−	+	−

Thus $\operatorname{Im} \phi(\xi) > 0$ for $\phi \in \Psi_{D_{\mathrm{CM}}}$. The alternating form on M,

$$E_\xi(x, y) = \operatorname{Tr}_{M/\mathbb{Q}}(\xi x \bar{y}),$$

satisfies $E_\xi(x, Jx) > 0$ for $x \neq 0$ and the Riemann bilinear relations. It is integral on $\mathcal{O}_M$; for any full lattice realizing this rational CM structure a positive integer multiple clears denominators. It therefore supplies a polarization, not necessarily a principal one. Its K_7 signature is $(3, 3)$ and $E_\xi(kx, ky) = \mathrm{N}_{K_7/\mathbb{Q}}(k) E_\xi(x, y)$.

Let $T(x, y) = \operatorname{Tr}_{M/K_7}(bx\bar{y})$. On the K_7-basis $1, u, \ldots, u^5$ its determinant is

$$\det T = \operatorname{disc}(f)\, \mathrm{N}_{L/\mathbb{Q}}(b) = (3^3 7^5)(-21^3) = -3^6 7^8 = -64827^2.$$

Indeed each u_a used above has norm 1 and $\mathrm{N}_{L/\mathbb{Q}}(\sqrt{21}) = -21^3$. The associated Hermitian form is

$$E_\xi(s_7 x, y) + s_7 E_\xi(x, y) = -14 T(x, y).$$

The sixth-power scalar is a K_7-norm, and 64827^2 is the norm of the rational element 64827. Hence the Hermitian discriminant class is -1 in $\mathbb{Q}^* / \mathrm{N}_{K_7/\mathbb{Q}}(K_7^*)$; clearing lattice denominators preserves this class. The identities and signs have distinct exact implementations in `experiments/fixed_field_cm_census_v98/k7_polarization_primary.py` and `independent/k7_split_polarization_v99.py`.

Markman's Theorem 1.5.1 applies to a polarized abelian sixfold of Weil type for an imaginary quadratic field, with Hermitian discriminant -1 [3, §1.1 and Thm. 1.5.1]. The constructed compatible polarization and balanced signature verify its hypotheses for D_{CM}. The theorem does not require a principal polarization. Consequently the two-dimensional rational Weil space of D_{CM} is algebraic. In particular it supplies both split lines of the relation

$$(1, -1, 1, -1, -1, -1)$$

in the six positive directions $1, 5, 11, 13, 17, 19$ of D_{CM}.

2.2.3 Five B/D_{CM} quartets from one source partition

For degree 42 Fermat factors use triples $(10, 12, 20)$ for B and $(1, 14, 27)$ for D_{CM}. The least-residue rule for $H^{1,0}$ in [10, eq. (2.1), p. 68] gives exactly $\Psi_B, \Psi_{D_{\mathrm{CM}}}$. The B triple has effective conductor 21 and the D_{CM} triple conductor 42; both have twelve character labels and the indicated six-dimensional admissible factors. Equality of marked CM types transports them to B, D_{CM} by complex isogenies.

The invariant support

$$S = \{(B, 1), (B, 19), (D_{\mathrm{CM}}, 31), (D_{\mathrm{CM}}, 41)\}$$

has its twelve Fermat residues partitioned into

$$(2, 10, 31, 41), \quad (12, 15, 18, 39), \quad (14, 28), \quad (20, 22).$$

For every unit modulo 42, each quadruple has least-residue sum 84; the pairs are opposite. These are blocks in Shioda's $\mathcal{B}_{42}^2$ and $\mathcal{B}_{42}^0$. The supportwise proof of Theorem 4.3, using Lemmas 4.1–4.2, makes the selected exterior character line algebraic [10, pp. 70–72]. No odd-degree hypothesis is imposed in these statements. This is not an application of the restricted hyperelliptic Theorem 5.4 at even degree.

The five B/D_{CM} quartets in the lattice basis below are precisely the translates $25S, 29S, 5S, 31S, S$. Their negatives are translates by -1. Thus all required quartet lines are algebraic. The complete codimension-two check, beyond these five needed inputs, is `independent/fermat_codim2_v100.py`; it independently verifies all $78 = 66 + 12$ invariant four-line supports and their source partitions.

2.3 From an integral relation basis to algebraic tensors

The following lemma isolates the unbounded argument. It concerns a particular collection of CM factors, not a general assertion that pairwise Hodge algebraicity settles arbitrary products.

Lemma 2.4 (Algebraic contractions)**.** *Let $F_1, \ldots, F_m$ be complex CM abelian varieties. Suppose their first cohomologies split into distinct nonzero weight lines within each factor, each with a unique opposite, over a number field E. Retain the factor labels and one representative of each opposite pair, and let $\mathcal{K}$ be the full integer lattice of relations among their synchronized Hodge characters. Suppose $\mathcal{K}$ has an integral generating set whose signed relation tensors, for both signs, are algebraic after extension to E. Suppose also that the embedding line projectors are induced by algebraic correspondences with coefficients in E. Then every rational Hodge class on every mixed power of the F_i is algebraic.*

Proof. First a known exterior generator can be moved to singleton H^1 slots. For a factor F of dimension d, the polynomial

$$\pi_1 = \prod_{\substack{0 \le k \le 2d \\ k \ne 1}} \frac{[n]^* - n^k \mathrm{Id}}{n - n^k}, \qquad n \ge 2,$$

is the rational H^1 projector, because $[n]^*$ acts as n^k in degree k. It is an algebraic correspondence. Pull a wedge of distinct embedding lines back along addition $F^q \to F$. Apply π_1 on each factor and the assigned embedding projector in each slot. Exactly one assignment survives, up to sign, yielding its pure Kunneth tensor on separate copies. For a generator on several original copies, perform this operation copy by copy. All classes remain in $\mathrm{Alg} \otimes E$.

Fix an ample rational divisor θ on F. On $Z \times F \times F$, diagonal pullback, cup product with θ^{d-1}, and pushforward along $Z \times F \to Z$ act on singleton entries by

$$\eta \otimes x \otimes y \longmapsto \eta \int_F x \wedge y \wedge \theta^{d-1}.$$

This is algebraic and lowers codimension by one. Hard Lefschetz and Poincare duality make the displayed alternating pairing nondegenerate. It is Hodge-group invariant. Distinct weights with

unique opposites therefore force a nonzero pairing on each opposite pair. The scalar lies in E and can be divided out. Only slots of the same marked factor are contracted. No inverse Lefschetz operator is used.

For $z \in \mathcal{K}$, write an exact integer combination of the known generators; replace a negative coefficient by copies of the known negative generator. Tensorize on fresh slots and take external products. If a direction occurs P times positively and N times negatively, contract $\min(P, N)$ opposite pairs. The surviving multiplicity is exactly the signed coordinate $P - N$ of z, and all contractions are nonzero. Rescaling gives the desired pure tensor in $\mathrm{Alg} \otimes E$. This permits nonconformal cancellation by correspondences, without squaring an exterior line or extracting a root of a cycle class.

Now take an invariant exterior monomial on a target mixed power. Set aside opposite pairs, algebraic by Lefschetz $(1,1)$; the remaining multiplicities give $z \in \mathcal{K}$. Construct its singleton tensor, reinsert the pairs by external products, and pull back along the product of diagonals assigning each slot to its intended target copy. The target labels are distinct, so this yields the required nonzero wedge up to sign. Temporary source powers need no separately assumed Hodge conjecture: the needed classes have been constructed on them.

The zero-weight exterior monomials span the rational Hodge space after splitting the CM torus. They all lie in $\mathrm{Alg} \otimes E$. The opposite inclusion holds because algebraic classes are Hodge; faithful scalar extension proves equality of the rational spaces, degree by degree. If a target factor is absent, its temporary slots have signed total zero and are all contracted away. A product with every exponent zero is a point. □

The required embedding projectors. For the present factors take a primitive element t in a full-rank order of actual CM endomorphisms. On H^1 the polynomial

$$p_\sigma = \prod_{\tau \neq \sigma} \frac{t^* - \tau(t)\mathrm{Id}}{\sigma(t) - \tau(t)}$$

projects onto the σ line. Compose it with π_1. The denominators are nonzero, and the result is an algebraic correspondence with coefficients in E; no model of the variety over E is asserted. For B, D_{CM} there are twelve embeddings of M; for Y there are six embeddings of $K = \mathbb{Q}(\zeta_7)$. Thus π_1 uses degrees 0–12 for B, D_{CM} and 0–6 for Y, and contraction uses respectively θ^5 and θ^2.

2.4 The synchronized nine-generator lattice

Give Y the K action $\zeta_7 \mapsto \rho^4$, where ρ is induced by $x \mapsto \zeta_7 x$. Its holomorphic type is $4\{1,2,3\} = \{1,4,5\}$ modulo 7. This is the relative marking used in the reviewed B/Y source. The sixfold census type $\{1,5,11,19,25,29\}$ is its inverse image modulo 42 and describes Y^2; it must not be counted as twelve distinct H^1 lines on one Y.

Use direction labels $(1,5,11,13,17,19)$ on B, D_{CM} and $(1,5,11)$ on Y, restricting the latter modulo 7 to $(1,5,4)$. For the *same* $g \in G = (\mathbb{Z}/42\mathbb{Z})^*$ put

$$w_{F,a}(g) = \begin{cases} +1, & ga \in \Psi_F, \\ -1, & ga \notin \Psi_F, \end{cases}$$

using modulus 42 for $F = B, D_{\mathrm{CM}}$ and modulus 7, with $\Psi_Y = \{1,4,5\}$, for Y. Let W be this 12×15 matrix. No independent Galois translation is applied to a single factor. Each factor's signed weights are distinct and nonzero, with a unique opposite.

The Hodge torus of this product is the smallest rational subtorus containing its Hodge circle. Equivalently, its cocharacter space is generated by the simultaneous Galois translates of the Hodge-circle cocharacter. Pairing an embedding character with these translates gives the entries $w_{F,a}(g)$, up to a common harmless sign convention. Consequently a signed multiplicity vector defines a trivial Hodge character exactly when it lies in $\ker_{\mathbb{Z}} W$. This uses the same g in every factor: separate translations would describe a different relative CM marking. Repeated copies repeat the same characters. In even cohomological degree, Hodge-circle invariance is precisely Hodge type (p,p), and splitting the torus over M therefore identifies its invariant exterior monomials with the scalar extension of rational Hodge classes.

The following rows are the full nine relation generators, in the column order $B(6) \mid D_{\mathrm{CM}}(6) \mid Y(3)$:

$$\left(\begin{array}{cccccc|cccccc|ccc}
0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & -1 & -1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\
0 & 1 & 1 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
1 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1
\end{array}\right) \tag{$\mathcal{R}$}$$

Lemma 2.5 (The full integer kernel)**.** *The rows in* ($\mathcal{R}$) *form a* $\mathbb{Z}$*-basis of* $\ker_{\mathbb{Z}} W$*.*

Proof. Order G increasingly. The last six rows of W are the negatives of the first six in reverse order. The first six rows, in one-based columns $1,2,3,4,5,7$, have determinant -32, so $\operatorname{rank} W = 6$. Every displayed vector is in its kernel. Project them to the one-based coordinates

$$6,8,9,10,11,12,13,14,15.$$

The resulting 9×9 integer matrix has determinant $+1$. These exact assertions, including an explicit integer inverse and full coordinate multiplication, are certified in `experiments/mixed_triple_v102/primary.py` and `experiments/mixed_triple_v102/primary.json`, with a separately reconstructed implementation and review listed in Section 2.6.

The nine vectors form a rational basis of the rank-nine kernel, and the indicated projection restricts to an isomorphism onto $\mathbb{Q}^9$. For an integral kernel vector, unimodularity gives integer coefficients from its projected coordinates. Subtracting this combination leaves a kernel vector with zero projection, hence zero. This proves equality with the full integer kernel, not just equal rational rank. □

Proof of Theorem 2.1. The first row of ($\mathcal{R}$) is the pure-D_{CM} Weil relation supplied by the split polarization and Markman's theorem. Rows two through six are the five B/D_{CM} quartet translates in Section 2.2. The last three are invariant B/Y quartets, algebraic by the one-product Shioda input. Both signs of all nine rows are algebraic. The exact weight checks give distinct nonzero weights and unique opposites within each factor, and the projectors above meet the remaining hypothesis of Lemma 2.4. Apply Lemma 2.5 and then Lemma 2.4.

In particular no separate theorem for $D_{\mathrm{CM}} \times Y$ is assumed. Its classes can be constructed using temporary B slots which contract away. Pairwise Hodge algebraicity alone would not prove this theorem; the full integral index-one statement is the extra input. □

2.5 The eight fixed-field types and individual powers

The exact CM-type census chooses one embedding from each of six conjugate pairs, so its complete domain has $2^6 = 64$ elements. There are eight Galois orbits, with the following representatives. Rank denotes the connected Hodge-group rank, not the Mumford–Tate rank.

Representative modulo42	Orbit	Rank	Weight structure
$1, 5, 11, 13, 17, 19$	12	5	one unit relation; B
$1, 5, 11, 13, 17, 23$	6	3	three independent pairs, multiplicity2
$1, 5, 11, 13, 19, 25$	12	5	one unit relation; D_{CM}
$1, 5, 11, 17, 23, 29$	12	6	six independent pairs
$1, 5, 11, 19, 25, 29$	6	3	three independent pairs, multiplicity2
$1, 5, 11, 23, 25, 29$	12	6	six independent pairs
$1, 11, 23, 25, 29, 37$	2	1	one pair, multiplicity6
$1, 13, 19, 25, 31, 37$	2	1	one pair, multiplicity6

The separate exact census implementations are `experiments/fixed_field_cm_census_v98/primary.py` and `experiments/fixed_field_cm_census_v98/independent.py`. Their retained outputs and hostile reviews bind the finite coverage, orbit data and character-relation calculations.

Proof of Theorem 2.3. For each of the six independent-weight orbits, an invariant exterior monomial on any power chooses equally many lines of each opposite weight. Pair them to write it as a product of degree-two invariant monomials. This argument includes repeated weight spaces and divisors mixing the copies. Lefschetz $(1,1)$ and rational scalar descent prove algebraicity.

For the two rank-five orbits the relation lattice of the six directions is generated respectively by

$$(1, -1, -1, 1, -1, 1), \qquad (1, -1, 1, -1, -1, -1).$$

Every coordinate is ± 1; rank five and a coordinate equal to one show that these are the full integral relation lattices. The two signs give the two split Weil lines on the original sixfold. They are algebraic on B by the J_{21} factor input, and on D_{CM} by the explicit K_7 polarization and Markman's theorem. Apply the one-factor case of Lemma 2.4, or equivalently distribute these six-line wedges among copies using addition and the embedding projectors.

Finally Galois relabeling of the M action and commensurability of the rank-one lattices transport the statement to every variety of a type in the indicated orbit. Positive integer scaling clears the integral polarization denominators without changing the relevant norm class. This proves the stated fixed-field, individual-powers scope. □

2.6 Certificate boundaries and retained extensions

The argument just given uses one pure Weil input and eight quartets. The shorter mixed-powers proof does not require the complete Graver basis or the subsequent cloud source searches. Those remain separate, reviewed results rather than implicit hypotheses.

Claim	Retained mathematical scope
C45	Individual powers of all eight rank-one M sixfold type orbits.
C46	All mixed powers of the three packet fibers, via B and J_7.
C47	Codimension two on all $B^r \times {D_{\mathrm{CM}}}^s$.
C48	All codimensions on one $B \times D_{\mathrm{CM}}$.
C49	Complete signed Graver-basis and finite generator reduction; the reduction alone asserts no new algebraicity.
C50	All mixed B/D_{CM} powers by an integral basis and contractions.
C51	All mixed $B/D_{\mathrm{CM}}/J_7$ powers, with the packet corollary above.
C52	All mixed products of the six degree-21 Fermat-type classes.
C53	All mixed products of all eight rank-one M sixfold classes.
C54	The stated full-degree CM-subfield factors and all their mixed products.

The V101 Graver computation gives 122 vectors up to sign, with degrees $4, 6, 8, 10$ and counts $6, 44, 24, 48$. Its completeness uses the separate signed-graph proof of a coordinate bound two and degree bound twelve. Of the 244 signed patterns, 168 repeat an embedding direction. The bounded source search produced independently verified Fermat witnesses for 144 of those signed patterns; the other 24 were not settled by that search. The contraction theorem establishes their algebraicity without changing those recorded search outcomes.

The following repository-relative paths locate the source-bound proof records. They are provenance references, not external assertions of specialist approval:

- `experiments/fixed_field_cm_census_v98/final_extension_exposition_review.json` and `experiments/fixed_field_cm_census_v98/k7_polarization_hostile_review.json`;
- `experiments/mixed_products_v100/second_primitive_review.json`, `experiments/mixed_products_v100/graph_review.json`, and `experiments/mixed_products_v100/second_primitive_codim3_review.json`;
- `experiments/mixed_powers_v101/graver_review.json`, `experiments/mixed_powers_v101/contraction_review.json`, and `experiments/mixed_powers_v101/source_results_review.json`;
- `experiments/mixed_triple_v102/primary.py`, `experiments/mixed_triple_v102/primary.json`, `independent/mixed_triple_v102.py`, `experiments/mixed_triple_v102/independent.json`, and `experiments/mixed_triple_v102/final_review.json`.

The independent V102 source and exact artifact hashes are bound by its comparison and final checkpoint reviews. Finite arithmetic comparisons support the stated integer identities; they do not replace the contraction, polarization, source-theorem or rational-descent arguments. No full fresh packet-certificate execution or consolidated release is claimed by these extension records.

3 Mixed products from six fixed-field CM-type orbits

Put $M = \mathbb{Q}(\zeta_{42}) = \mathbb{Q}(\zeta_{21})$ and $G = (\mathbb{Z}/42\mathbb{Z})^\times$. As in Section 2, types refer to rational first cohomology: an embedding label g belongs to the type when its eigenline lies in $H^{1,0}$. We use the canonical types Ψ_i in Table 1, retaining their indices in the eight-orbit census. These are abstract field markings, not additional geometric CM points on the McMullen curve.

Theorem 3.1 (Six-orbit mixed products)**.** *Let $X_1, \dots, X_N$ be complex abelian sixfolds, each with an embedding $M \hookrightarrow \mathrm{End}^0(X_j)$ and $\dim_M H^1(X_j, \mathbb{Q}) = 1$. Suppose each cohomological CM type is a Galois translate of one of the six types in Table 1. Then every rational Hodge class on $X_1 \times \cdots \times X_N$ is algebraic, in every codimension. The factors may repeat, and their lattices may be chosen independently. The empty product is allowed.*

i	Ψ_i	$t \pmod{21}$	q	$\dim A_t$	e	h
0	$\{1,5,11,13,17,19\}$	$(1,1,19)$	21	6	1	19
1	$\{1,5,11,13,17,23\}$	$(1,7,13)$	21	6	1	11
4	$\{1,5,11,19,25,29\}$	$(3,3,15)$	7	3	2	11
5	$\{1,5,11,23,25,29\}$	$(1,6,14)$	21	6	1	1
6	$\{1,11,23,25,29,37\}$	$(1,4,16)$	21	6	1	1
7	$\{1,13,19,25,31,37\}$	$(7,7,7)$	3	1	6	1

Table 1: Degree-21 Fermat presentations. Here q is the character order, $e = [M : \mathbb{Q}(\zeta_q)]$, and h transports the induced type to Ψ_i. An admissible factor A_t need not be simple.

Proof. Let F_m be the smooth plane Fermat curve. Its Jacobian has admissible rational character factors indexed by Galois orbits of triples $t = (t_1, t_2, t_3)$ with nonzero entries and sum zero modulo m. If t has order q, its factor A_t has dimension $\varphi(q)/2$ and rank-one rational first cohomology over $E = \mathbb{Q}(\zeta_q)$; no simplicity assumption is required [10, §2, eqs. (2.2)–(2.4) and Definition 2.1]. We specify the holomorphic convention used for these factors. On $x^m + y^m = 1$, the differentials

$$x^{a-1} y^{b-m}\, dx, \qquad a, b \geq 1, \quad a + b \leq m - 1,$$

are regular and have diagonal character $(a, b, m-a-b)$. At $y = 0$, use $dx = -y^{m-1}x^{1-m}dy$; at infinity their local factor has exponent $m - a - b - 1 \geq 0$. Their distinct characters and number $(m-1)(m-2)/2$ give a holomorphic basis. Thus the holomorphic labels of t are exactly those u for which $\sum_j \langle u t_j \rangle_m = m$, where residues lie in $\{1, \dots, m-1\}$, in agreement with [10, eq. (2.1), p. 68].

For each row of Table 1, set $d = \gcd(21, t_1, t_2, t_3)$, $q = 21/d$, and $v = t/d$. The ordered character v has image μ_q. Its factor therefore has the E-action just described, with holomorphic type

$$T_q(v) = \left\{ u \in (\mathbb{Z}/q\mathbb{Z})^\times : \sum_j \langle u v_j \rangle_q = q \right\}.$$

The induced M-type is

$$T = \{ g \in G : g \bmod q \in T_q(v) \} = \left\{ g \in G : \sum_j \langle g t_j \rangle_{21} = 21 \right\}.$$

Direct residue calculations give $hT = \Psi_i$ in each displayed row. For example the embedding fibers have size $e = 12/\varphi(q)$, so the dimensions and copy multiplicities in the table follow without identifying an elliptic or three-dimensional factor with a sixfold. The type-stabilizer orders in row order are $1, 2, 2, 1, 6, 6$; in particular several types are induced.

We explain why induction gives an actual field action on A_t^e. Write $V = H^1(A_t, \mathbb{Q})$. Choosing an E-basis of M identifies

$$H^1(A_t^e, \mathbb{Q}) = V^e \simeq M \otimes_E V.$$

Multiplication on the first factor is a faithful rank-one M-action. It preserves the Hodge decomposition: after splitting E, a matrix over E only mixes copies of the same embedding component, whose Hodge type is homogeneous. The line labeled g lies over the E-embedding labeled $g \bmod q$, so its type is exactly T.

These rational Hodge endomorphisms are rational algebraic endomorphisms. Indeed, dualizing a rational Hodge map to homology gives a real linear map commuting with the complex structures. Clearing denominators makes it a map of lattices, hence a holomorphic homomorphism of complex tori. Its graph is algebraic for abelian varieties. The reversal of composition under pullback causes no problem for the commutative field M. Thus the construction supplies an embedding $M \hookrightarrow \mathrm{End}^0(A_t^e)$, not just a choice of split eigenvalues. Re-mark this action by $\iota_{\mathrm{new}} = \iota_{\mathrm{old}} \circ \sigma_h$. An old label g becomes gh, since $\sigma_g(\sigma_h(a)) = \sigma_{gh}(a)$; hence the new type is hT.

Two rank-one rational M Hodge structures with the same type are M-linearly Hodge-isomorphic: an M-linear isomorphism preserves each embedding line and its Hodge bidegree. The weight-one lattice argument above turns such an isomorphism into a complex algebraic isogeny after clearing denominators. This works for induced types as well as simple ones. A further field re-marking realizes any Galois translate. Consequently each X_j is complex-isogenous to the corresponding A_t^e. Every finite product in the theorem is therefore a degree-21 Fermat-type abelian variety in Shioda's sense, up to isogeny.

The remaining input is the full semigroup condition P_{21}. The sentence preceding the published Theorem 3.3 of da Silva, with its Appendix computation, establishes P_m for $m = 21, 27$ [2]. We use this published result as a literature input, without claiming an independent replay of its computation. The superscript in the accompanying hypersurfaces X_m^n denotes dimension, not Cartesian power. Shioda's Theorem 4.4 states that P_m^{6c-2} implies rational Hodge algebraicity in codimension c for every degree-m Fermat-type abelian variety [10, Theorem 4.4, p. 72]. Full P_{21} supplies this hypothesis for every $c \geq 1$. It applies to the identified product, including repeated admissible factors; codimension zero is immediate. This use of Theorem 4.4 supplies the correspondence and multiplicity statement directly, without inferring mixed algebraicity from separate factors. Finally, isogeny pullback and pushforward preserve cycle classes, with $f_* f^* = (\deg f)\mathrm{id}$, and transfer the conclusion to the original product over $\mathbb{Q}$. □

The finite identifications, including character kernels, embedding fibers, and action transporters, are retained in `experiments/six_orbit_v109/identification.json`; the human proof and its primary-source hypotheses remain essential. Theorem 3.1 concerns the six listed isogeny classes. The two remaining census types, 2 and 3, are addressed in Section 4. The six-class theorem imposes no principal polarization and makes no claim of a new literal McMullen point or prescribed moduli normalization. The equality $\mathbb{Q}(\zeta_{42}) = \mathbb{Q}(\zeta_{21})$ does not by itself lower the order of a degree-42 Fermat character.

4 Arbitrary mixed products for the fixed degree-twelve field

The six-class result and two additional Fermat-character cycles extend algebraicity to arbitrary mixed products for the fixed field $M = \mathbb{Q}(\zeta_{42})$. This field restriction is essential.

Theorem 4.1 (All mixed products for the fixed field)**.** *Let $X_1, \ldots, X_N$ be complex abelian sixfolds equipped with embeddings $M \hookrightarrow \mathrm{End}^0(X_j)$ such that $\dim_M H^1(X_j, \mathbb{Q}) = 1$ for every j. Then every rational Hodge class on $\prod_j X_j$ is algebraic, in every codimension. Repetitions, arbitrary choices of lattices and the empty product are allowed.*

Choose representatives X_i for the eight canonical CM-type orbits. Besides the six types in Table 1, these are

$$\Psi_2 = \{1, 5, 11, 13, 19, 25\} = \Psi_{D_{\mathrm{CM}}}, \qquad \Psi_3 = \{1, 5, 11, 17, 23, 29\}.$$

The eight orbits exhaust the $2^6 = 64$ choices of one embedding from each conjugate pair; their respective sizes are $12, 6, 12, 12, 6, 12, 2, 2$. The marked weight-one isogeny argument in the proof of Theorem 3.1 reduces the theorem to arbitrary powers of these eight representatives, including induced types.

4.1 Actual embedding lines and an integral relation certificate

Set $R = (1, 5, 11, 13, 17, 19)$ and order $G = (\mathbb{Z}/42\mathbb{Z})^\times$ increasingly. For every factor use the *same* row $g \in G$ and set

$$w_{i,a}(g) = 21_{\Psi_i}(ga) - 1.$$

The Hodge-torus argument of Section 2.4 shows that zero sums of these columns are exactly the invariant signed tensor characters. All twelve actual M-embedding lines on each X_i remain distinct even if their Hodge-torus weights coincide. There are 48 actual conjugate-pair coordinates (i, a), $a \in R$.

Retain the following representatives for distinct pairs of opposite weight columns, with the factors ordered by i and the labels by size:

i	0	1	2	3	4	5	6	7
R_i	R	$\{1, 5, 11\}$	R	R	$\{1, 5, 11\}$	R	$\{1\}$	$\{1\}$.

Let W be the resulting 12×32 matrix and $e_{i,a}$ its coordinate vectors. Every original column is uniquely a signed retained column of the same factor. Thus there is a surjection $P : \mathbb{Z}^{48} \to \mathbb{Z}^{32}$ and a section J including the retained actual labels, with $PJ = 1$ and $W_{\mathrm{act}} = WP$. Consequently

$$\ker_{\mathbb{Z}} W_{\mathrm{act}} = J(\ker_{\mathbb{Z}} W) \oplus \ker P.$$

The sixteen generators of $\ker P$ have the form $f_{i,a} - \epsilon f_{i,b}$, where a is not retained and $w_{i,a} = \epsilon w_{i,b}$. Their tensors use two singleton copies: $L_{i,a} \otimes L_{i,-b}$ if $\epsilon = 1$, and $L_{i,a} \otimes L_{i,b}$ if $\epsilon = -1$. They are split components of rational degree-two Hodge classes and are algebraic by Lefschetz $(1,1)$. This justifies the reduction without identifying different geometric eigenlines or assuming they are interchanged by a prescribed endomorphism.

Here is a precise finite certificate for the remaining integer kernel. All coordinate indices in this paragraph are *zero-based*. The pivot columns of W are $(0, 1, 2, 3, 4, 6)$ and its rank is six. Write F for the increasing list of the other 26 indices. The unique kernel chart K whose rows in F form the identity has integral entries. Thus $\ker_{\mathbb{Z}} W = K\mathbb{Z}^{26}$. For $S = \{0, 1, 4, 5, 6, 7\}$, use the same column order on W_S; its analogous free-coordinate kernel chart has fourteen columns. Embed these fourteen columns in $\mathbb{Z}^{32}$, and follow them by the nine vectors in $(\mathcal{R})$, embedding B, D_{CM}, Y respectively in factors $0, 2, 4$ with the displayed retained labels. Call this ordered list of 23 known vectors $\mathcal{B}$. They have rank 19 and are algebraic by Theorems 3.1 and 2.1; here X_4 is isogenous to Y^2.

Define two further relations

$$q = -e_{0,1} - e_{0,17} + e_{1,1} + e_{2,1},$$
$$r = e_{0,1} - e_{0,5} + e_{0,13} - 2e_{1,1} + e_{3,1}.$$

For $h \in G$, send each label a to ha and then reduce to its signed representative using P. Order the distinct resulting integer vectors lexicographically in the fixed 32 coordinates. Each of q and r has twelve such translates. Form a square matrix by taking, in order, the projections to F of:

(i) the first nineteen vectors of $\mathcal{B}$ (the fourteen S-chart columns followed by the first five rows of ($\mathcal{R}$));

(ii) the lexicographically first translate of q;

(iii) the first six lexicographically ordered translates of r.

Its determinant is -1. Its retained integer inverse, multiplied by the corresponding full 32-coordinate matrix, gives exactly K. These explicitly specified finite identities are certified in `experiments/all_eight_v109/lattice.json`, reconstructed by `experiments/all_eight_v109/review/independent_check.py`, and regressed in `tests/test_v109_reductions.py`. The chart identity proves integral surjectivity onto the free coordinates, while unimodularity proves that the known vectors and the two seed orbits generate the full integer kernel. Equal rational rank alone is not used to infer saturation.

4.2 Two explicit algebraic source tensors

Each occurrence below has its own marked source copy. A negative sign in a relation means the conjugate embedding; the coefficient -2 in r requires two distinct copies of X_1. The following source presentations have exactly the indicated canonical cohomological types, with nonzero entries and character order 42:

relation	factor	Fermat triple
q	0	$(1, 1, 40)$
q, r	1	$(1, 13, 28)$
q	2	$(1, 14, 27)$
r	0	$(1, 20, 21)$
r	3	$(1, 6, 35)$.

Indeed $\sum_j \langle gt_j \rangle_{42} = 42$ selects Ψ_i for each row. Each associated rational character factor has rank-one M first cohomology and dimension six; the type-1 factor need not be simple. The marked isogeny argument above identifies it with the required X_i. The source embeddings for q are $(0, 41), (0, 25), (1, 1), (2, 1)$; those for r are $(0, 1), (0, 37), (0, 13), (1, 41), (1, 41), (3, 1)$, with the repeated pair occupying different copies.

Concatenate the four triples after their indicated multipliers for q. The resulting multiset has the following partition:

$$(1, 41), \quad (1, 41), \quad (2, 13, 25, 25, 27, 34), \quad (14, 28).$$

For the six source copies of r a partition is

$$(1, 41), \quad (1, 41), \quad (6, 20, 21, 37),$$
$$(8, 21, 26, 29), \quad (13, 29), \quad (14, 14, 21, 35).$$

Every block C is balanced: $\sum_{a \in C} \langle ga \rangle_{42} = 21|C|$ for every $g \in G$. The only length-six block splits into the two nonzero zero-sum triples $(2, 13, 27)$ and $(25, 25, 34)$. Thus the blocks satisfy exactly Shioda's length-two and length-four conditions and, for length six, his additional product-of-triples condition [10, Theorem 4.3 and Lemmas 4.1–4.2, pp. 70–72]. That theorem has no odd-degree

restriction; it applies at level 42. We use its proof supportwise for these two characters, rather than asserting that every level-42 invariant support has such a partition. The permitted block eigenspaces are algebraic, and the algebraic induction correspondences combine them into the concatenated line.

The algebraic correspondences in those lemmas transfer the supported Fermat-character cycles to the nonzero tensor components of the marked curve products. For q the concatenation has twelve entries, hence Fermat dimension ten and Hodge type $(5,5)$; the corresponding curve tensor has type $(2,2)$. For r the Fermat dimension is sixteen and the curve tensor has cohomological degree six, with types $(8,8)$ and $(3,3)$ respectively. These are the required Tate shifts, not degree-preserving identifications. Passing through Jacobian H^1, the rational character projectors, and the marked isogenies gives the desired nonzero singleton-copy tensors on the X_i. All projectors are algebraic correspondences with coefficients in the splitting field M. Distinct source copies keep repeated characters in different exterior positions. Nonzero transfer follows from the finite-degree dominant maps in Lemma 4.2: their pullback of a marked summand is a sum of products of the curve-character lines, and pushforward divided by the map degree recovers the original class. The split line and the algebraic subspace are M-defined; inclusion after extension to $\mathbb{C}$ descends faithfully to M. An individual nonrational eigenline is therefore not being called a rational cycle. The exact multiset, all twelve balance tests, actual CM types and the six-block splitting are independently checked by `experiments/all_eight_v109/validate_witness.py`.

4.3 Integral combinations, contractions and rational descent

Proof of Theorem 4.1. Work temporarily with cycle-class coefficients extended from $\mathbb{Q}$ to M. Rational polynomials in multiplication-by-n pullback isolate H^1; actual M-endomorphisms then give the individual embedding-line projectors. They distinguish all actual lines, even when their Hodge weights agree. Addition pullback and these projectors separate any source wedge into a specified nonzero singleton-copy tensor. Conversely diagonal pullback assembles singleton factors into any legitimate target exterior monomial.

We need nonzero contractions for actual conjugate embedding pairs. Choose an ample rational divisor θ on each sixfold X_i. On two singleton copies the operation

$$(x,y) \longmapsto \int_{X_i} x \smile y \smile \theta^5$$

is algebraic: diagonal pullback, cup product and proper pushforward. It is nonzero on $L_{i,a} \otimes L_{i,-a}$ by the polarization's Riemann bilinear positivity applied to v and $\overline{v}$. This remains valid for induced types with repeated weights; nondegeneracy of the whole pairing alone would not establish that individual assertion. No inverse-Lefschetz correspondence or principal polarization is needed.

Suppose algebraic singleton tensors represent signed relations v_j and their negatives. For an integer expression $z = \sum_j n_j v_j$, take external products, choosing $-v_j$ when $n_j < 0$. In each actual coordinate contract all opposite pairs. The surviving tensor has the signed multiplicities of z and is a nonzero scalar multiple of its canonical tensor. It remains algebraic with coefficients in M. This proves closure under arbitrary signed integer combinations, without a conformal decomposition or extraction of roots of cycle classes.

The known classes and the two source tensors therefore give their Galois translates. Here coefficient Galois action preserves the rational cycle-class image and translates all embedding labels simultaneously. An actual translate need not equal J of its reduced translate, but their difference lies in $\ker P$. The sixteen divisor relations and the contraction argument correct this difference.

The integer certificate then supplies every reduced kernel relation, and the split exact sequence supplies every actual kernel relation.

Finally take an invariant exterior monomial on any requested product. Remove matched occurrences of opposite actual labels, accounting for them by divisor classes. The remaining signed vector belongs to $\ker_{\mathbb{Z}} W_{\mathrm{act}}$ and hence has an algebraic singleton tensor. Restore the removed pairs and assemble the tensor into the target copies by diagonals. No duplicate identical embedding appears within one target exterior factor, so the result is nonzero. Auxiliary source types absent from the target are contracted away completely. All invariant split monomials are thus algebraic. They span the extension to M of the rational Hodge space; faithful scalar extension implies equality of its rational subspace with the rational cycle-class image. Complex isogeny invariance gives the stated arbitrary factors. □

This conclusion is limited to rank-one first cohomology over the fixed field M. It neither asserts the Hodge conjecture for all CM fields nor identifies any new geometric McMullen point, polarized isomorphism, or principally polarized moduli normalization.

Corollary 4.2 (CM factors with fields contained in the fixed field)**.** *Let A be a complex abelian variety. Suppose that every simple factor B in a complex isogeny decomposition of A admits an embedding $E \hookrightarrow \mathrm{End}^0(B)$ of a CM field with $[E : \mathbb{Q}] = 2\dim B$, and that E admits a $\mathbb{Q}$-field embedding into $M = \mathbb{Q}(\zeta_{21})$. Then every rational Hodge class on A is algebraic. The same holds for arbitrary finite mixed products of varieties satisfying this hypothesis, including all their powers.*

Proof. Write $A \sim \prod_j B_j^{m_j}$ over $\mathbb{C}$, and identify each chosen E_j with its image in M. Set $e_j = [M : E_j]$ and $X_j = B_j^{e_j}$. Then $\dim X_j = 6$. If $V_j = H^1(B_j, \mathbb{Q})$, the full-degree CM action makes V_j rank one over E_j, and an E_j-basis of M identifies

$$H^1(X_j, \mathbb{Q}) = V_j^{e_j} \simeq M \otimes_{E_j} V_j.$$

Multiplication on the first factor gives a faithful rank-one M-action preserving the Hodge decomposition. Its type is the inverse image of the E_j-type under restriction of embeddings. The weight-one lattice argument in the proof of Theorem 3.1 realizes this action in $\mathrm{End}^0(X_j)$; maximal-order integrality on the original lattice is not required. Thus every X_j meets Theorem 4.1.

Choose N_j with $e_j N_j \geq m_j$. Theorem 4.1 gives rational Hodge algebraicity on $Z = \prod_j X_j^{N_j} = \prod_j B_j^{e_j N_j}$. Projection onto $A' = \prod_j B_j^{m_j}$ has an algebraic section inserting origins in the extra factors. For a Hodge class α on A', pullback to Z is algebraic; pullback along the section recovers α and preserves algebraic cycle classes. Complex isogeny invariance transfers this conclusion to A. A mixed product of varieties satisfying the hypothesis has simple factors satisfying the same hypothesis, so the argument also proves the last assertion. We use the specific mixed-product theorem above, not a general closure of the Hodge conjecture under products. □

Corollary 4.3 (An intrinsic splitting-field criterion)**.** *Let $M = \mathbb{Q}(\zeta_{21})$ and let A be a complex abelian variety. The following conditions are equivalent:*

1. $\mathrm{MT}(H_1(A, \mathbb{Q}))$ *is a $\mathbb{Q}$-torus split over M;*
2. *every simple isogeny factor B of A admits a unital embedding of a CM field E into $\mathrm{End}^0(B)$, with $[E : \mathbb{Q}] = 2\dim B$, and a $\mathbb{Q}$-field embedding $E \hookrightarrow M$.*

Every rational Hodge class on an abelian variety satisfying these conditions is algebraic. The same holds for every finite mixed product of such varieties, including all their powers.

Proof. We use homology covariantly: $H_1(B,\mathbb{Q})$ is a polarizable Hodge structure of weight -1 and type $(-1,0),(0,-1)$. The equivalence between these rational Hodge structures and complex abelian varieties up to isogeny identifies their endomorphisms; see [4, Theorem 6.8 and Corollary 6.9].

Suppose first that (1) holds. For a simple factor B, put $V = H_1(B,\mathbb{Q})$ and let T be the image of $\mathrm{MT}(A)$ on V. This is a torus split over M and contains the Hodge homomorphism. The rational Hodge structure V is simple: a proper nonzero Hodge substructure would give a proper abelian subfactor up to isogeny. Consequently V is an irreducible rational representation of T.

Let $R \subset \mathrm{End}_{\mathbb{Q}}(V)$ be the rational algebra generated by this representation. Equivalently, its scalar extension contains the geometric image of T; rational points need not be used. Over M, distinct characters decompose V_M into weight spaces, and R_M is the full algebra of scalars on those distinct spaces. Indeed, distinct characters are linearly independent as regular functions, so their evaluations span the corresponding diagonal algebra. Thus R is finite étale and commutative. Its invariant rational subspaces are exactly those of T. Since V is a faithful simple R-module, R is a field E and V has dimension one over E. Therefore $[E:\mathbb{Q}] = 2\dim B$, while $E \otimes_{\mathbb{Q}} M \simeq M^{[E:\mathbb{Q}]}$ gives an embedding $E \hookrightarrow M$. The algebra E commutes with the Hodge homomorphism, so the covariant Hodge equivalence embeds it into $\mathrm{End}^0(B)$.

It remains to see that E is CM. As a subfield of the abelian Galois extension $M/\mathbb{Q}$, it is preserved by complex conjugation. If conjugation acted trivially, E would be totally real. The complex structure J on $V_{\mathbf{R}}$ commutes with E, and rank one would then give

$$J \in \mathrm{End}_{E\otimes\mathbf{R}}(V_{\mathbf{R}}) = E \otimes \mathbf{R} \simeq \mathbf{R}^{[E:\mathbb{Q}]},$$

contradicting $J^2 = -1$. Conjugation is therefore nontrivial; its fixed field is totally real and E is a totally imaginary quadratic extension of that field. This proves (2).

Conversely, the field in (2) makes $H_1(B,\mathbb{Q})$ rank one over E. The Hodge homomorphism lies in the centralizer of E, namely $\mathrm{Res}_{E/\mathbb{Q}}\mathbf{G}_m$. Minimality of the Mumford–Tate group puts $\mathrm{MT}(B)$ inside this torus. The Mumford–Tate group is connected: the connected Hodge torus lands in its identity component, and minimality excludes further components. Thus it is a subtorus. Since $M/\mathbb{Q}$ is Galois and contains E, every embedding of E takes values in M, so this restriction-of-scalars torus, and each of its subtori, splits over M. The Mumford–Tate group of a finite direct sum embeds into the product of the individual groups. It is consequently a subtorus split over M; equality with that product is unnecessary. Isogenies and repetitions preserve the conclusion, proving (1).

Corollary 4.2 now gives Hodge algebraicity. The same subtorus argument proves that arbitrary finite mixed products again satisfy (1), giving the last assertion. □

Here the Mumford–Tate group must already be a torus over $\mathbb{Q}$. Splitting over M is an additional condition on that torus, not a claim that a noncommutative group becomes a torus after extending scalars.

5 Scope and further questions

The field is fixed throughout. No additional CM point on a curve is constructed, and isogeny invariance does not identify principal polarizations. The useful continuation is a precise comparison of the explicit source cycles and saturated relation presentations with the classical Fermat results and broader CM constructions. A future general statement would require its own algebraic inputs and integral generation proof, rather than extrapolation from this finite character census.

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